

Cong Cao

PhD Student in Computer Vision

Abu Dhabi, UAE | cong.cao@mbzuai.ac.ae

[Google Scholar](#) | [GitHub](#)

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

PhD in Computer Vision

Advisor: **Prof. Hao Li**

📅 Aug. 2024 – Present

📍 Abu Dhabi, UAE

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

MSc in Computer Vision

Advisor: **Prof. Hao Li**

📅 Aug. 2022 – Jun. 2024

📍 Abu Dhabi, UAE

University of Bristol

MSc in Robotics | Graduated with Distinction

Advisor: **Dr. Paul O'Dowd**

📅 Sep. 2021 – Nov. 2022

📍 Bristol, UK

Xidian University

BEng in Intelligence Science and Technology

GPA: 3.8/4.0

📅 Sep. 2018 – Jul. 2022

📍 Xi'an, China

RESEARCH EXPERIENCE

3D Garment Generation and Virtual Try-on

- Reusing 3D garments across diverse characters is challenging when transfer methods rely on manual fitting, shared rigs, or predefined correspondences. We developed LUIVITON, an automated virtual try-on system that combines SMPL-based correspondence and registration with neural cloth simulation to fit rest-posed garments to posed humanoids. [\[SIGGRAPH 2026 / TOG\]](#)
- Diverse panel shapes and layouts make sewing-pattern generation challenging; autoregressive generation can accumulate errors, while discrete representations hinder differentiable optimization. We designed an implicit panel representation and trained a latent flow-matching model to learn how panels combine into sewing patterns. The learned prior supports pattern generation and completion, while panel differentiability enables refitting across body shapes. [\[SIGGRAPH 2026\]](#)

PUBLICATIONS

01 Learned Universal Interoperable Virtual Try-on

Cong Cao, Xianhang Cheng, Jingyuan Liu, Yujian Zheng, Zhenhui Lin, Ren Li, Meriem Chkir, Hao Li.

ACM Transactions on Graphics 45(4), Article 156, 2026 (SIGGRAPH 2026). [Paper](#)

02 Learning Sewing Patterns via Latent Flow Matching of Implicit Fields

Cong Cao*, Ren Li*, Corentin Dumery, Hao Li.

ACM SIGGRAPH 2026 Conference Papers, Article 179, 2026. [Paper](#)

PUBLICATIONS CONTINUED

03 Spatio-Temporal Garment Reconstruction Using Diffusion Mapping via Pattern Coordinates

Yingxuan You*, Ren Li*, Corentin Dumery, **Cong Cao**, Hao Li, Pascal Fua.

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), accepted 2026. [Paper](#)

04 Single View Garment Reconstruction Using Diffusion Mapping Via Pattern Coordinates

Ren Li, **Cong Cao**, Corentin Dumery, Yingxuan You, Hao Li, Pascal Fua.

ACM SIGGRAPH 2025 Conference Papers, 2025. [Paper](#)

* Equal contribution.

INDUSTRY EXPERIENCE

Yalla Group Limited

📅 May 2023 – Jul. 2023

Research and Development Intern

📍 Dubai, UAE

- Fine-tuned CLIP on annotated real-world chatroom images to build an image classifier for content filtering.

TECHNICAL EXPERTISE

Programming and tools: Python, PyTorch, Marvelous Designer, Blender.

TEACHING EXPERIENCE

Teaching Assistant, MBZUAI

CV804 3D Geometry Processing

📅 Jan. – May 2026

CV802 Advanced 3D Computer Vision

📅 Aug. – Dec. 2025

HONORS AND AWARDS

MBZUAI Fully Funded Scholarship

📅 2024 – Present

MBZUAI Fully Funded Scholarship

📅 2022 – 2024

Diligence Scholarship

📅 2020

Annual University Scholarship

📅 2019